

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Regular Examination, 2021

URP 2115
(Urban Planning Techniques)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) How planning is related to the development? (05)
(b) Mention some penalties for violation of acts and rules mentioned in Master Plan. Also describe the preparation and execution process of Master Plan. (15)
(c) Illustrate your concept about the DMDP structure plan. What propositions were made for agricultural, flood control and transportation sector in DMDP structure plan? (15)
2. (a) What are the national and specific urban strategies targeted in strategic plan for Bangladesh? (10)
(b) How was the selection and execution done in Detail Area Planning (DAP) zones for Khulna City? (10)
(c) What are the major functions of Local Plans? Make a tabulated comparison among District Plan, Action Area Plan and Subject Plan. (15)
3. (a) What are the major Participatory Area Planning techniques? (05)
(b) Discuss about the Guided Land Development (GLD) process. Why it has not been applied in Bangladesh yet? (15)
(c) Where Land Readjustment Plan (LRP) has been applied yet? Discuss about the sectors that can be benefitted by LRP. (15)
4. (a) Why is the application of Land Banking techniques difficult for Dhaka city? (05)
(c) Mention the purpose and process of operation of Land Banking. (10)
(c) Illustrate the features and characteristics of the Area Improvement Plan (AIP). (10)
(d) Discuss about the process of operation about the Land Readjustment Plan (LRP). (10)

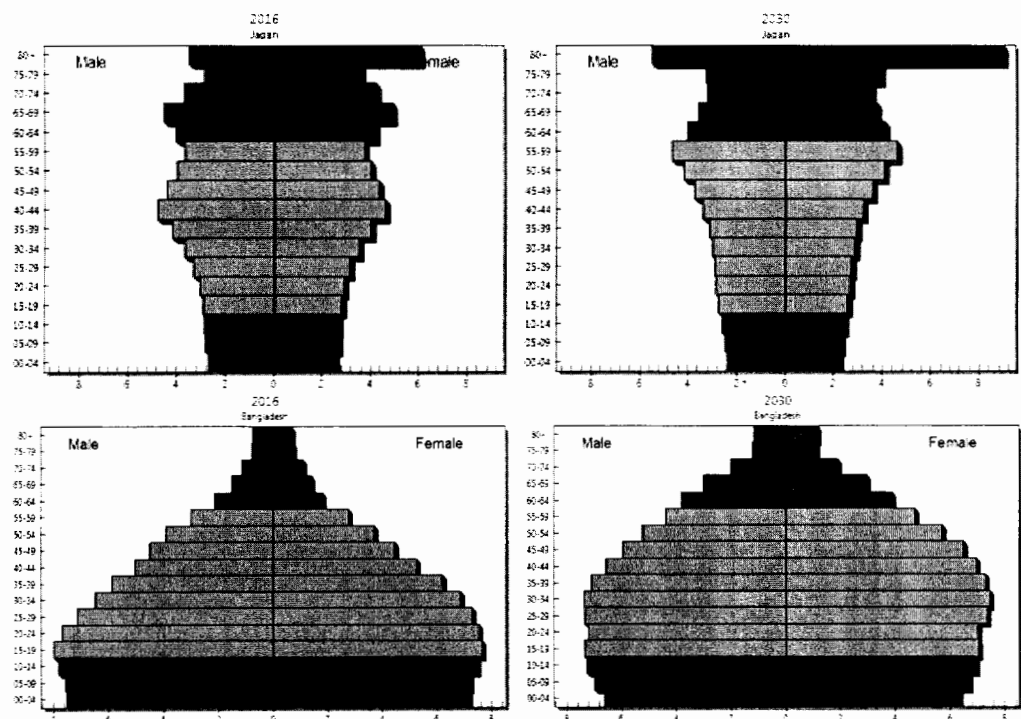
Section – B

5. (a) Explain the methods that can be used to identify basic and non-basic sectors. (15)
(b) What is Locational Quotient (LQ) approach? What are the assumptions of LQ approach? Using LQ approach, find out the basic and non-basic sectors of the following economic industries. (20)

SIC code	Industry	Local	National
A	Agricultural	5240	70,96,533
C	Construction	1378	4642735
D	Food production	1687	16,014
I	Hotels and Lodgings	124	1,33,4210
H	Local Government	4060	9,32,6667
Total		12,489	22,41,6159

Further mention the limitations of LQ approach with examples.

6. (a) Elaborate on the current structure of Bangladesh population in terms of total (10)
number, age distribution, and male/female ratio.
- (b) Below is a population projection of Bangladesh and Japan. What is your (15)
understanding about the future population composition and its economic
effects on Bangladesh and Japan?



- (c) Explain the following Alternative Extrapolation Curves based on their merits, (10)
demerits and use cases:
- i. Linear Curve
 - ii. Logistic Curve
7. (a) Calculate the total new housing units required for a population of 59250. (15)

Assumption	Value
Group Home	
Proportion of persons living in Group Homes	3.11%
Housing Mix	
Single Family	75.36%
Multi-Family	24.64%
Household Size	
Single Family	2.7
Multi-Family	1.9
Vacancy Rate	
Single Family	4.52%
Multi-Family	12.75%

- (b) Provide an example of quantitative technique of parks, recreation and (10)
open-space (RPOS) needs assessment.
- (c) "Homelessness is a major issue and requires to be attacked from different (10)
angles." -- Explain some of these frontiers for tackling the problem.
8. (a) Explain the following: (15)
- i. Growth Area Fixation Approach
 - ii. Landuse Delineation Approach
 - iii. Urban Renewal
- (b) Explain Suitability Analysis with the help of a proper example demonstrating (20)
the calculations that take place.

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Regular Examination, 2021

URP 2113
(Site and Area Planning)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) List out the various elements of a Site Inventory Checklist. (10)
(b) Describe elevation, slope and aspect. What does the 7% slope in a NE direction entail? (10)
(c) Explain the various structural and drainage characteristics. (15)
2. (a) Explain the guiding principles of Recreation, Parks and Open Space (RPOS) planning and its challenges. (10)
(b) What are the dependences of vertical curve? Draw the components of a horizontal curve and label them. (10)
(c) Given that the radius of a horizontal curve is 5m and the intersection/central angle (Δ) is 60° . Calculate the following: (10)
(i) Length of Tangent
(ii) Long Chord
(iii) Middle Ordinate
(iv) Length of Arc
(v) External Distance
(d) Find the void ratio with the following data: $V_A=5\text{m}^3$, $V_W=2000000\text{cm}^3$ and $V_S=10\text{m}^3$ (05)
3. (a) Describe the site selection process with an appropriate example. (15)
(b) Elaborate “SMART” growth in the context of site planning. (10)
(c) Prepare a wind rose diagram with assumed data and provide necessary measures that can be taken for proper wind flow. (10)
4. (a) Diagrammatically explain the following: (10)
(i) Burgess Zone Model
(ii) Hoyt Sector Model
(b) Position different land use controls from passive to active. (10)
(b) Why is it important to calculate surface runoff in site and area planning? (15)

Section – B

5. (a) What do you understand by site planning? Write down the scope of site planning. (15)
(b) “Neighbourhood is a nucleus for the development of the local social life of the environment” – Explain the statement with relations to site and area planning. (10)
(c) Write down the good features of a planned neighborhood. (10)
6. Narrate the following pattern of subdivision (any five): (35)
(i) Gridiron pattern;
(ii) Modified grid;
(iii) Curvilinear pattern;
(iv) Rectilinear pattern;
(v) Cul-de-sac pattern;
(vi) Loop street pattern;
(vii) Cluster pattern

7. (a) What are the available technologies for site and area planning? Write about the benefits and limitations of using technology for site and area planning in the modern era. (15)
- (b) What are the general conditions for the approval of a private residential housing project under the Private Residential Project Land Development Rules 2012? (20)
8. Write down the objectives, performance criteria and prescriptive measures for the following elements of residential subdivision. (any five) (35)
- (i) Density;
 - (ii) Setback;
 - (iii) Solar Access;
 - (iv) Landscaping;
 - (v) Soil and Water Management;
 - (vi) Access way Design;
 - (vii) Waste Minimization and Management

Khulna University of Engineering & Technology
Department of Urban & Regional Planning
BURP 2th Year 1st Term Examination, 2021.
URP 2191
(Statistics for Planner - I)

Full Marks: 120

Time: 3.0 hrs

- N.B.** i) Figures in the right margin indicate full marks.
ii) Answer any **two** questions from each section in separate script.
iii) Normal distribution table, z-table
iv) Graph paper

Section – A

1. (a) Define statistics and mention the limitations of statistics. (10)
(b) What are the sources of data? Write down the major steps of a questionnaire survey and the key characteristics of a question. (10)
(c) Draw Histogram, Frequency Polygon and Ogive curve from the following data: (15)

Class	10-15	15-20	20-25	25-30	30-35	35-40	40-45
Frequency	5	10	15	8	17	9	3

2. (a) What are the common measures of central tendency? Explain 67-95-99 rule of a normal distribution. (10)
(b) What are the advantages and disadvantages of Arithmetic mean? Write short notes on alternatives for arithmetic mean with example. (12)
(c) Calculate the arithmetic mean and median of the frequency distribution of CGPA of 60 students given below. (13)

CGPA	No. of Students
2.80-2.99	8
3.0 – 3.19	15
3.20 – 3.39	10
3.40 – 3.59	17
3.60 – 3.79	8
3.80 – 3.99	2

3. (a) What are the common measures of data variability? Explain with example that data showing apparent central tendency actually can have variability. (13)
(b) Explain the difference between moment about mean and moment about arbitrary value. (10)
(c) Calculate (i) Q_3 (3rd quartile) (ii) D_7 (7th decile) and P_{75} (75th percentile) from the following data:- (12)

Class	Frequency
13-15	10
15-17	16
17-19	32
19-21	60
21-23	28
23-25	17
25-27	9

4. (a) Write short notes on the following: (15)
(i) Cumulative frequency; (ii) Pie chart vs bar chart; (iii) Histogram and trend line.
(b) Graphically explain the concepts of Normality, Skewness and Kurtosis. (12)
(c) Prove that central moment (around mean) is always 0. (08)

Section – B

5. (a) Define the following terms: (08)
(i) Population,
(ii) Sample,
(iii) Parameter,
(iv) Statistic.
- (b) Discuss the types of variables in statistics with examples. (10)
- (c) A student experiences difficulties with malfunctioning alarm clock. Instead of using one, she decides to use three. What is the probability that at least one of his alarm clocks works correctly if each individual alarm clock has a 98% chance of working correctly? (07)
- (d) Make a comparative overview of nominal, ordinal, interval and ratio scale data. (10)
6. (a) Discuss statistically dependent vs independent events with necessary examples. (08)
- (b) According to a survey, the probability that a family owns a car if its monthly household income is greater than BDT 100,000 is 0.80. Of the household surveyed, 58% had incomes over BDT 100,000 and 50% had a car. A family is randomly selected; find out the probability that the family has a car and an income over BDT 100,000 monthly. (10)
- (c) You take a trip by air that involves three independent flights. If there is a 75% chance each specific leg of the trip is on time, what is the probability all three flights arrive on time. (07)
- (d) A committee of 6 person is to be formed from 8 men and 5 women for a Town Planning Board. In how many ways can this be done when
(i) at least 3 women are included? (10)
(ii) at least 2 women are included?
7. (a) Write down the advantages and shortcoming of Relative Frequency Probability. (10)
- (b) The probability that a management trainee will remain with a company is 0.60. The probability that an employee earns more than BDT. 50,000 per month is 0.50. The probability than employee is a management trainee who remained with the company or who earns more than BDT. 50,000 per month is 0.70. What is the probability that an employee earns more than BDT. 50,000 per month, given that he is a management trainee who stayed in the company? (12)
- (c) Jar 1 contains 6 green, 4 black balls; Jar 2 contains 4 green, 6 black balls and Jar 3 contains 5 green, 5 black balls. Randomly selected a Jar and a ball is drawn from it. If the ball drawn is Green, find the probability that it is drawn from the first Jar. (13)
8. (a) What is mean by probability distribution? Discuss types of probability distribution with examples. (10)
- (b) The incidence of occupational disease in an industry is such that the workers have a 20% chance of suffering from it. What is the probability that out of 6 workers, 4 or more will contract disease? (13)
- (c) Automobiles arrive at the exit of Khan Jahan Ali Bridge at the rate of two per minute. The distribution of arrivals approximates a Poisson distribution. (12)
(i) What is the probability that no automobiles arrives in a particular minute?
(ii) What is the probability that at least one automobiles arrives during a particular minute?

Khulna University of Engineering & Technology
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CE 2107
(Construction Technology and Materials)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume suitable data if necessary.

Section – A

1. (a) What is brick? Give the constituents of brick clay and state their functions. (10)
(b) Draw the flow chart of manufacturing of brick and discuss the preparation of clay in brick manufacturing. (15)
(c) Draw the plan of Bull's trench kiln with neat sketches and explain also its working principle. (10)
2. (a) Define aggregate. Discuss the classification of aggregate according to size. (06)
(b) What is grading? Sieve analysis data of two varieties of sands, one from Sylhet and another from Kushtia are given below. Weight of each sample is 500 gm. Calculate the amount of each variety of sand in a mixture of 20 kg to obtain the combined fineness modulus of 2.50. (20)

Sieve Size	Sylhet sand (wt. retained in gm)	Kushtia sand (wt. retained in gm)
No. 4	0	0
No. 8	61	14
No. 16	110	35
No. 30	114	52
No. 50	120	262
No. 100	93	135
No. 200	2	2

- (c) Draw the qualitative gradation curve for well graded, gap graded and uniformly graded sample and discuss briefly. (09)
3. (a) What is cement? Write the mineral constituents of ordinary portland cement and discuss briefly the function of each in cement. (10)
(b) Write short notes on: (12)
 - (i) Initial setting time and final setting time.
 - (ii) False setting and flash setting of cement.
 - (iii) Hydration and soundness of cement.
- (c) Explain the different steps in the dry process of manufacturing portland cement with neat flow diagram. (13)
4. (a) Define concrete. What are the functions of water in concrete? (08)
(b) Write short notes on: (12)
 - (i) Workability of concrete
 - (ii) Slump test
 - (iii) Segregation
 - (iv) Bleeding
- (c) Discuss the effect of water content on workability. (05)
(d) State the influence of time and temperature on workability. (10)

Section – B

5. (a) Define deep foundation with suitable examples. What is the difference between footing and foundation? (11)
(b) Draw the neat sketch of the following foundations. Also describe why and when they are best suited for - (12)
 - (i) Combined footing
 - (ii) Mat foundation
- (c) Write short notes on – (12)
 - (i) Earthen cofferdam
 - (ii) Crib cofferdam
 - (iii) Cellular cofferdam
6. (a) Write down the comparative merits and demerits of English and Flemish bond. (10)
(b) Define voissoirs, skew-back and Haunch of the arch. Classify arches according to their shape. (12)
(c) What are the components of flooring? What functional requirements should be considered for the selection of suitable type of floor construction? (13)
7. (a) What is cavity wall? What are the advantages of cavity wall? (10)
(b) Write down the position of cavity wall and discuss about constructional details. (13)
(c) Write down short notes on: (12)
 - (i) Double-hung windows
 - (ii) Semi-circular arch
 - (iii) Bearing joints
8. (a) What is stair? What are the requirements of a good stair? Describe briefly. (12)
(b) Classify stair with their neat sketch. (10)
(c) What is plastering? What are the requirements of good plaster? (08)
(d) Is it possible to plaster over paint? Give suitable explanation. (05)

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Hum 2117
(Sociology for Planners)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What do you mean by 'Sociology'? (05)
(b) Discuss in brief the origin and scope of Sociology. (15)
(c) What is community and society? Distinguish between society and community. (15)
2. (a) What is family? (05)
(b) Explain the functions and importance of family. (15)
(c) What is Association and Institutions? Explain with examples. (15)
3. (a) What do you mean by culture? (05)
(b) Mention the characteristics of culture. Discuss the elements of culture. (15)
(c) What is cultural lag? Discuss briefly the history and culture of Bangladesh. (15)
4. (a) What is the meaning of 'social stratification'? (05)
(b) Explain the major types of social stratification with appropriate example. (20)
(b) Discuss the types of marriage. (10)

Section – B

5. (a) Define industrialization. (05)
(b) What are the trends of industrialization in Bangladesh? (15)
(c) What are the impacts of industrialization on society and environment? Discuss in brief. (15)
6. (a) Differentiate between pre-industrial and industrial cities. (10)
(b) What is Primate city? Explain the basic criteria of Mega city and Meta city. (10)
(c) Critically evaluate Homer Hoyt's Sector theory of city development. (15)
7. (a) Differentiate between collective poverty and cyclical poverty in short. (05)
(b) How does parenting style affect juvenile delinquency? Explain citing example from Bangladesh. (15)
(c) Explain the role of the agencies of social control with examples. (15)
8. (a) Why is social stratification universal? Explain the different forms of social stratification with example. (10)
(b) What are the impacts of population growth on environment? Discuss in brief. (10)
(c) Explain the social structure of Bangladesh after the liberation war. (15)